

STAT 516: Some discrete distributions

Lecture 11: Discrete Uniform, Binomial, Negative Binomial, and Geometric Distributions

Prof. Michael Levine

February 18, 2021

Discrete uniform distribution

- ▶ The pmf is $P(X = x) = \frac{1}{n}$ for any $x = 1, 2, \dots, n$
- ▶ Note that the cdf is $F(k) = \frac{k}{n}$ for any $1 \leq k \leq n$
- ▶ The expected value is

$$\mu = E(X) = \sum_{x=1}^n xp(x) = \frac{n+1}{2}$$

Additional moments of the discrete uniform distribution

► Similarly,

$$E(X^2) = \sum_{x=1}^n x^2 p(x) = \frac{1}{n} \sum_{x=1}^n x^2$$
$$\frac{1}{n} \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$$

► Thus,

$$\sigma^2 = V(X) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2 - 1}{12}$$

Summary

- For a discrete uniform distribution on $\{1, 2, \dots, n\}$

$$E(X) = \frac{n+1}{2}; V(X) = \frac{n^2-1}{12}; E(X - \mu)^3 = 0$$

Bernoulli distribution: a model

- ▶ A **Bernoulli experiment** is the basic unit
- ▶ The two outcomes must be mutually exclusive and exhaustive
- ▶ The pmf for X that has Bernoulli distribution is

$$p(x) = p^x(1 - p)^{1-x}$$

where $x \in \{0, 1\}$

- ▶ Thus, $\mathbb{E} X = p$ and $\text{Var}(X) = \sigma^2 = p(1 - p)$

Binomial distribution: a model

- ▶ A number of successes in a sequence of n **Bernoulli trials** X has the **binomial distribution**
- ▶ If n is fixed, the number of ways to choose x successes out of n is $\binom{n}{x} = \frac{n!}{x!(n-x)!}$
- ▶ The **binomial pmf** of X is

$$p(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

for $x = 0, 1, 2, \dots, n$ and 0 elsewhere

- ▶ Verify directly that $p(x)$ is a true pmf
- ▶ n and p are the **parameters** of the binomial distribution

An alternative way of working with a binomial distribution

- ▶ Represent $X = \sum_{i=1}^n X_i$ where each $X_i \sim b(1, p)$ and independent
- ▶ Check that the expectation and the variance can be easily obtained this way!

Example I

- ▶ A multiple-choice test with 20 questions has five possible answers for each question. A student answers each question at random and independently.
- ▶ X is the number of questions a student answers correctly.
- ▶ Then, $X \sim \text{Bin}(n, p)$ with $p = \frac{1}{5} = 0.2$ and $n = 20$

Example I



$$\begin{aligned} P(\text{ The student gets every answer wrong }) \\ = P(X = 0) = .8^2 = 0.0115 \end{aligned}$$



$$\begin{aligned} P(\text{ The student gets every answer right }) \\ = P(X = 20) = .2^{20} = 1.05 \times 10^{-14} \end{aligned}$$

- ▶ In order to pass the test, a student has to answer 13 questions right;

$$P(\text{ The student will pass }) = \sum_{x=13}^{20} \binom{20}{x} .2^x .8^{20-x} = 0.000015$$

- ▶ You can use R commands $\text{dbinom}(k, n, p)$ for the pmf $P(X = k)$ and $\text{pbinom}(k, n, p)$ for the cdf $P(X \leq k)$

Example II

- ▶ A fair coin is tossed $n = 2m$ times. What is the probability that the number of heads obtained will be an even number?
- ▶ The number of heads $X \sim \text{Bin}(2m, 0.5)$ and

$$\begin{aligned} P(X = 0) + P(X = 2) + \cdots + P(X = 2m) &= \sum_{x=0}^m \binom{2m}{2x} \frac{1}{2^{2m}} \\ &= \frac{2^{2m-1}}{2^{2m}} = \frac{1}{2} \end{aligned}$$

because $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}$

Summary

► Let $X \sim \text{Bin}(n, p)$. Then

1. $\mu = E(X) = np$; $\sigma^2 = \text{Var}(X) = np(1 - p)$
2. $E[(X - \mu)^3] = np(1 - 3p + 2p^2)$
3. $E[(X - \mu)^4] = np(1 - p)[1 + 3(n - 2)p(1 - p)]$

Corollary

- ▶ Let $\beta = \beta(n, p)$ and $\gamma = \gamma(n, p)$ be the skewness and kurtosis of $X \sim \text{Bin}(n, p)$. Then, $\beta, \gamma \rightarrow 0$ as $n \rightarrow \infty$.
- ▶ Thus, no matter what $0 < p < 1$ is chosen, for large n the binomial distribution becomes nearly symmetric and *normal*-like as n gets large

Geometric and Negative Binomial Distributions

- ▶ A couple will have children until they have had two boys; X is the number of children they will have
- ▶ $X \sim NB(r, p)$ with $r = 2$ and $p = 0.5$ where

$$p(x) = P(X = x) = (x - 1)(.5)^x$$

for $x = 2, 3, \dots$

- ▶ Then e.g.

$$\begin{aligned} P(\text{ Couple will have at least one girl }) &= P(X \geq 3) \\ &= 1 - P(X = 2) = 1 - 0.25 = 0.75 \end{aligned}$$

- ▶ Surprisingly, $P(X \geq 6) = 1 - P(X \leq 5) \approx 0.19$

Example

- ▶ The number of people you expect to meet who share the same birthday is $X \sim \text{Geo}(p)$ with $p = \frac{1}{365}$
- ▶ The pmf of X is $p(x) = p(1 - p)^{x-1}$
- ▶ Thus,

$$P(X > k) = \sum_{x=k+1}^{\infty} p(1 - p)^{x-1} = p \sum_{x=k}^{\infty} (1 - p)^x = (1 - p)^k$$

- ▶ For illustration, $P(X > 1000) = \left(\frac{364}{365}\right)^{1000} = 0.064$

An important property of the geometric distribution

- ▶ Geometric distribution is **memoryless**: for any two positive integers m and n

$$P(X > m + n | X > n) = P(X > m)$$

- ▶ We will see later that this property is shared with the exponential distribution which is the continuous analog of the geometric distribution!

Some basic facts about the geometric and negative binomial distributions

- ▶ Geometric X : $\mathbb{E} X = \frac{1}{p}$, $\text{Var } X = \frac{1-p}{p^2}$
- ▶ Negative binomial X : $\mathbb{E} X = \frac{r}{p}$, $\text{Var } X = \frac{r(1-p)}{p^2}$
- ▶ Note that the result for the negative binomial can be easily obtained by using the following representation:

$$X = X_1 + X_2 + \cdots + X_r$$

where X_i is the geometric random variable measuring the number of additional trials needed to obtain i th success after $i - 1$ success was achieved

Example

- ▶ Capture-recapture sampling is a classical example
- ▶ The population consists of N animals; D captured, tagged and released
- ▶ In the second sample size n , the number of tagged $X \sim \text{Hypergeo}(n, D, N)$
- ▶ Typically, the true population size is estimated as $N = \frac{nD}{X}$
- ▶ Can be problematic in practice if $X = 0$ or if the tagged animals cluster together after release
- ▶ Corrected version called *Petersen's estimator* is used in wildlife biology, when taking a census, by governments for estimating tax fraud and/or the number of people afflicted with a particular infection

Basic properties of the hypergeometric distribution

- ▶ Check that its $\mathbb{E} X = n \frac{D}{N}$ and $\text{Var} X = n \frac{D}{N} \frac{N-D}{N} \frac{N-n}{N-1}$
- ▶ The expectation is the same as that of the binomial distribution with $p = \frac{D}{N}$
- ▶ The variance is different from the binomial by the factor $\frac{N-n}{N-1}$ which is called a **population correction factor**
- ▶ Can show that if $\frac{D}{N} \rightarrow p$ for some $0 < p < 1$ as $N \rightarrow \infty$ the hypergeometric pmf

$$p(x) = \frac{\binom{N-D}{n-x} \binom{D}{x}}{\binom{N}{n}} \rightarrow \binom{n}{x} p^x (1-p)^{n-x}$$